AI Value

Digital & AI Transformation

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INSTALLATION GUIDE — MINIFIED RUNTIME — CUSTOMER SIDE — LINUX

CortexSkills

Install the CortexSkills production runtime on your Linux server (private cloud or on-premises), without Docker, with the reference data of every vertical and SMEs, and activate your licence

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AI Value

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1. What you install

CortexSkills is delivered as a single compressed file, Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz: a folder to extract on your server, which holds the production runtime. It runs with Node.js 22 installed on the server; no build and no Internet access are needed. Your data does not leave your servers: the database, attachments and backups are in the data/ folder. Nothing is sent to the vendor.

Item	Role
cortexskills	The launcher: start, admin (installation console), secrets, backup, restore, version.
installer-service.sh	Installs the Linux service (systemd): dedicated system user, folder rights, start at boot.
app/	The application, read-only for the service.
data/	Database, attachments, daily backups and installed licence. This is the only data to protect.
.env	Your settings: address, secrets, administrator, mail (created from env.example).
config/	cortexskills.service (hardened systemd unit) and nginx-cortexskills.conf (HTTPS).
licence/	Where you put the licence file received from the vendor.
LISEZMOI.txt · README.txt	Reminder of the essential commands, in French and English.

What the package contains — and what it does not

The production runtime: the minified server (file by file, without comments, internal names shortened), the compiled and minified web client (split per page, pre-compressed with Brotli and gzip, without source maps) and the libraries needed at run time.

The vendor public key: it verifies the licence, it cannot create one.

No readable source code, no test, no source map, no signing tool, no private key.

Node.js is not included: it is a prerequisite of the machine (Node.js 22 LTS, 22.13 or newer; validated version: 22.22.2). This keeps the package to a single compressed file.

Run-time optimisations: only the screens opened are loaded (the first display loads about 370 KB of JavaScript instead of 2.2 MB), static files are served pre-compressed and cached by the browser for a year (their name changes with each version), and Node.js keeps the compiled code between two starts (data/.node-compile-cache).

1.1 The reference data

At first start, the installation loads 61 reference organizations: 32 large companies and 29 SMEs, in every vertical (29 sectors and a universal case). Each has two complete projects (Digital and AI focus) with its processes, tasks, steps, questionnaires, plans, documents and indicators: they are examples to look at, in English, French and Arabic.

No demonstration account

The sign-in page names no account; no demonstration password exists in this production version.

The people of the reference organizations cannot sign in: their accounts are disabled and their password, drawn at random at installation, is known to nobody.

Only the platform administrator (ADMIN_EMAIL) signs in at first; they look at the reference organizations read-only (Portfolio › Groups and organizations). Your own users are those of your organization (chapter 8).

They have no licence: they do not count in the number of users of your licence. For an empty database, set REFERENCE_DATA=non in .env before the first start.

2. Prerequisites

2.1 Server

Item	Minimum	Recommended (300 users and more)
System	Linux 64-bit x86-64 with glibc 2.28 or newer	Ubuntu 22.04 / 24.04 LTS, Debian 12, RHEL / Rocky / AlmaLinux 9
Node.js	22 LTS, 22.13 or newer (chapter 3)	22.22.2
Processor	2 vCPU	4 vCPU
Memory	4 GB	8 GB
Disk	20 GB free: application (about 80 MB), database with the reference data (about 0.9 GB), daily backups (about 0.8 GB each, 7 days by default)	50 GB on backed-up storage
Software	systemd; tar and xz; sha256sum; an HTTPS reverse proxy (Nginx recommended)	Nginx and Certbot (Let's Encrypt) or your company's certificate

Alpine Linux (musl) and arm64 processors are not supported by this package: use the Docker package, or ask the vendor for the matching version.

2.2 Network

Need	Detail
DNS name	For example skills.your-company.com, pointing to the server (an internal DNS is enough for an intranet).
Inbound	443/TCP for the users (and 80/TCP to obtain the Let's Encrypt certificate). Port 4000 stays closed: the application only listens on 127.0.0.1.
Installation	No Internet access needed if Node.js is installed from the official archive (isolated server possible).
Outbound in operation (optional)	Your SMTP server (often 587/TCP); graph.facebook.com on 443/TCP for WhatsApp Business.
Browsers	Recent Chrome, Edge, Firefox or Safari.

2.3 What you receive from the vendor

File	Role
Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz	The runtime: a single compressed file (xz) with the minified application, the launcher, the service, the Nginx example, the settings template and the list of checksums.
Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz.sha256	The SHA-256 checksum of the file, to check it is intact.
Installation_Guide_Runtime_Customer_Linux_CortexSkills_EN.docx / .pdf	This guide.
User_Guide_Customer_CortexSkills_EN.docx / .pdf	The User Guide: use of the application on a complete real case.
Later: <Your_company>.lic	Your licence file, sent by the vendor after your request (chapter 9).

3. Install Node.js 22

Node.js 22 LTS (22.13 or newer) must be installed on the server. Two methods; then check with node -v.

Method 1 — official archive (also works without Internet on the server). Download node-v22.22.2-linux-x64.tar.xz and SHASUMS256.txt from https://nodejs.org/dist/v22.22.2/ (on a connected workstation, then copy them to the server):

```
grep " node-v22.22.2-linux-x64.tar.xz$" SHASUMS256.txt | sha256sum -c -
sudo tar -xJf node-v22.22.2-linux-x64.tar.xz -C /usr/local --strip-components=1
node -v          # v22.22.2
```

Method 2 — NodeSource repository (server connected to the Internet).

```
# Ubuntu, Debian
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash - && sudo apt-get install -y nodejs
# RHEL, Rocky, AlmaLinux
curl -fsSL https://rpm.nodesource.com/setup_22.x | sudo bash - && sudo dnf install -y nodejs
node -v          # v22.x (22.13 or newer)
```

The nodejs package of the standard Ubuntu or Debian repositories is too old (18 or 20): do not use “apt install nodejs” without NodeSource. If Node.js is installed elsewhere, give its path in the CORTEXSKILLS_NODE variable (for example /opt/node22/bin/node).

4. Check and extract the package

As administrator (sudo). The recommended folder is /opt/cortexskills: the systemd service protects the home folders (/home), do not install the application there.

```
cd ~                                    # folder holding the files received
sha256sum -c Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz.sha256          # must answer: Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz: OK
sudo mkdir -p /opt/cortexskills
sudo tar -xJf Runtime_Customer_Linux_CortexSkills_1.15.1.tar.xz -C /opt/cortexskills --strip-components=1
cd /opt/cortexskills && sha256sum -c --quiet SHA256SUMS.txt && echo "content intact"
./cortexskills version
```

The last command shows “CortexSkills 1.15.1 · customer edition · Node.js 22.22.2 · data /opt/cortexskills/data”. If a check fails, do not install: ask the vendor for a new copy.

5. Settings (.env)

```
sudo cp env.example .env
./cortexskills secrets          # shows JWT_SECRET and CHANNEL_SECRET_KEY (and a third value, not used here)
sudo nano .env
```

Setting	Value	Example
PUBLIC_URL	The address users type, in HTTPS; it appears in the links sent by e-mail and WhatsApp.	https://skills.your-company.com
CORS_ORIGIN	The same address.	https://skills.your-company.com
HOST, PORT	Listening interface and port. 127.0.0.1: only the reverse proxy (or the local browser) reaches it.	127.0.0.1, 4000
JWT_SECRET	Random value of 32 characters or more (signs the sessions).	shown by “cortexskills secrets”
CHANNEL_SECRET_KEY	Another random value (encrypts the stored credentials).	shown by “cortexskills secrets”
ADMIN_EMAIL, ADMIN_NAME, ADMIN_PASSWORD	The platform administrator, created at first start. Password: 12 characters min., upper case, lower case, digit, symbol. Clear ADMIN_PASSWORD afterwards.	it.ops@your-company.com
REFERENCE_DATA	First start: “oui” loads the reference organizations; “non” creates an empty database.	oui
DATA_DIR	Optional: data folder. By default data/ in the package folder.	/var/lib/cortexskills
BACKUP_RETENTION_DAYS	Days the daily backups are kept (about 0.8 GB each).	7
SMTP_*	Your mail server for invitations, reminders and notifications.	smtp.your-company.com, 587
WHATSAPP_*	Optional: WhatsApp Business Cloud API.	—

The package enforces the on-premises mode (signed licence required) and hides all commercial information, whatever the value of DEPLOYMENT_MODE or COMMERCIAL_INFO in .env: these settings are ignored.

Start-up checks

The application refuses to start while a setting is dangerous — short or default secret, two identical secrets, address not in HTTPS, open CORS, vendor private key present — and lists each problem in its log.

Keep a copy of .env in your vault: changing JWT_SECRET signs every user out; losing CHANNEL_SECRET_KEY makes the stored mail credentials unreadable.

6. Install the service and start

```
cd /opt/cortexskills
sudo ./installer-service.sh
journalctl -u cortexskills -f       # Ctrl+C to quit
```

installer-service.sh checks Node.js (22.13 or newer), creates the “cortexskills” system user (no shell, no password), makes app/ read-only (owned by root), gives data/ to the service only, restricts .env to root and the service, installs /etc/systemd/system/cortexskills.service with the path of Node.js, then starts the service and enables it at server start.

At first start, the application creates the production database — reference catalog, templates, platform administrator and reference organizations (1.1) — then starts. This step takes 5 to 15 minutes depending on the server: do not interrupt it. The log shows “Production database ready … 61 reference organizations (no account can sign in)”, then “CortexSkills API is running”. Health check, on the server:

```
curl -s http://127.0.0.1:4000/api/health
{"status":"ok","initialized":true,"mode":"onprem","version":"1.15.1","demoAccounts":fal
se}
```

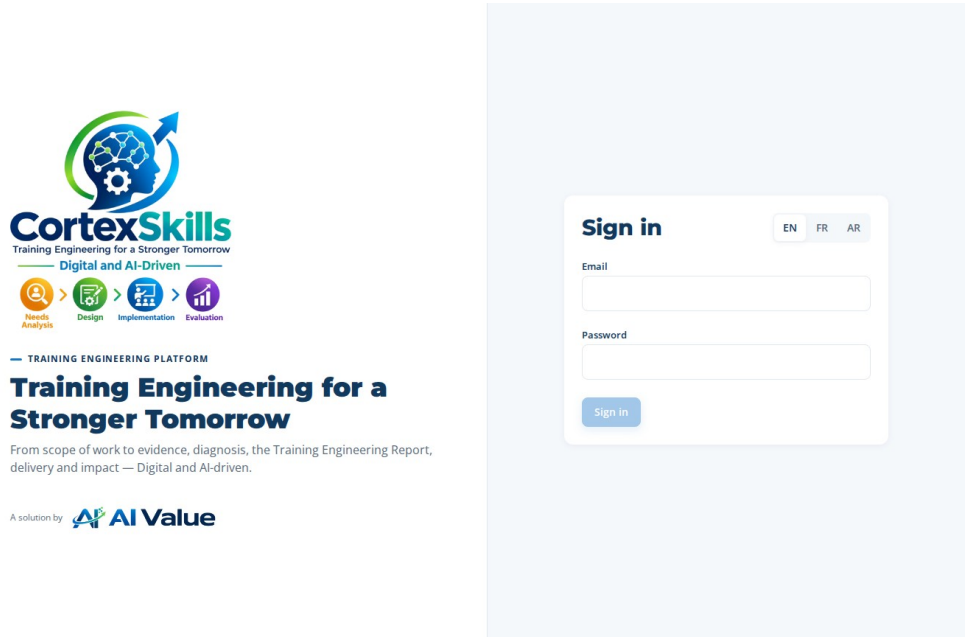
Then clear the value of ADMIN_PASSWORD in .env (sudo nano .env), then sudo systemctl restart cortexskills.

Service protection (config/cortexskills.service)	Effect
User=cortexskills, NoNewPrivileges	The process has no privilege and cannot obtain any.
ProtectSystem=strict, ReadWritePaths=…/data	The whole file system is read-only for the service, except data/.
ProtectHome, PrivateTmp, PrivateDevices	No access to home folders, to the others’ /tmp, or to devices.
UMask=0077	The files created (database, backups, attachments) are readable by the service only.
TimeoutStartSec=30min	Leaves time for the first start to load the reference data.

7. HTTPS (Nginx)

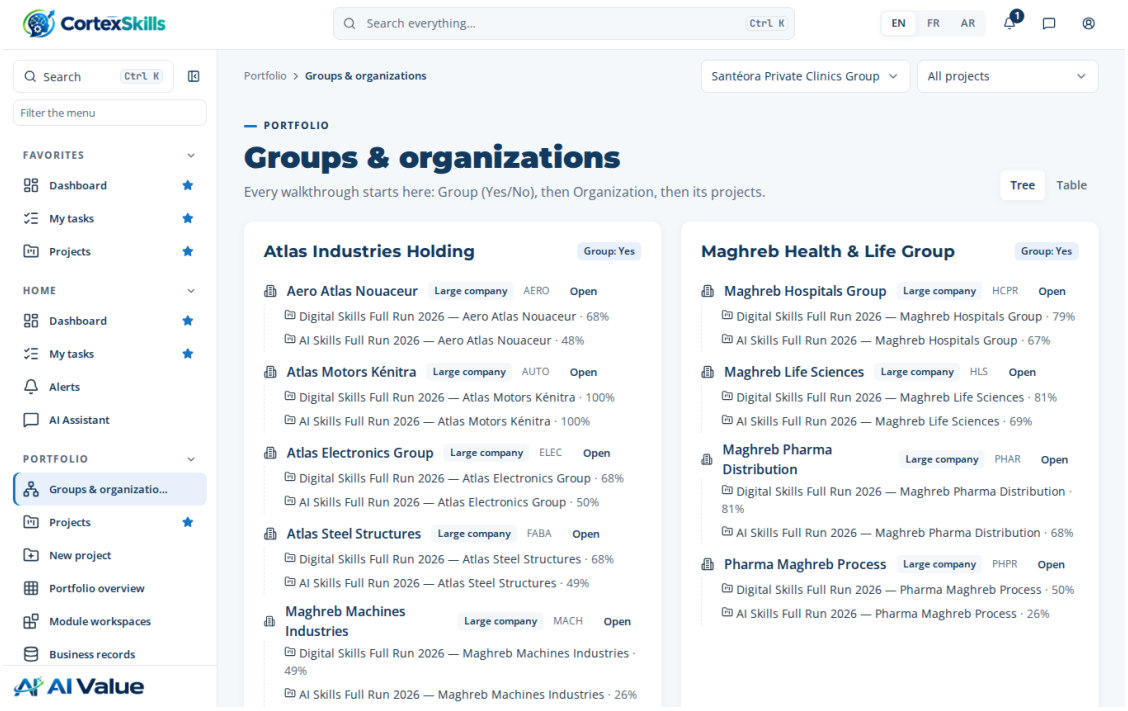
```
sudo apt -y install nginx certbot python3-certbot-nginx
sudo cp config/nginx-cortexskills.conf /etc/nginx/conf.d/cortexskills.conf
sudo nano /etc/nginx/conf.d/cortexskills.conf      # your DNS name and your certificate
sudo certbot certonly --nginx -d skills.your-company.com
sudo nginx -t && sudo systemctl reload nginx
```

The file redirects port 80 to HTTPS, accepts only TLS 1.2 and 1.3, allows 30 MB attachments and forwards to http://127.0.0.1:4000 with the X-Forwarded-* headers. Another reverse proxy (Apache, HAProxy, F5, IIS ARR, Traefik) works the same way. Then open your address:



The sign-in page at your address: no demonstration account

Sign in with ADMIN_EMAIL and its password: the reference organizations are in Portfolio › Groups and organizations.



The reference organizations, seen by the platform administrator

7.1 The reference organizations

Sector	Large companies	SMEs
AEC & Construction (AEC)	Oasis Construction & Engineering	Souss BTP Solutions
Aerospace & Defense (AERO)	Aero Atlas Nouaceur	Precision Aero Composites
Automotive (AUTO)	Atlas Motors Kénitra	Tanger Auto Parts
Consumer Packaged Goods (CPG)	Sahel Consumer Brands	Savonnerie Beldi
Consumer Products Extended (CPEX)	Sahel Home & Living	Poterie Safi Design
Distribution (DIST)	Horizon Distribution	Distri Oriental
Electronics & High-Tech (ELEC)	Atlas Electronics Group	Circuitec Oujda
Energy & Utilities (ENUT)	Oasis Energy & Utilities	Solaris Maintenance Services
Fabrication & Assembly (FABA)	Atlas Steel Structures	Atelier Métal Berrechid
Food & Beverage (FNB)	Sahel Agro Industries	Coopérative Laitière Al Baraka
Footwear & Apparel (FTAP)	Atlas Textile & Footwear	Atelier Cuir Fès
Healthcare & Life Sciences (HLS)	Maghreb Life Sciences	BioLab Diagnostics Rabat
Healthcare Providers (HCPR)	Maghreb Hospitals Group ; Santéora Private Clinics Group ; Regional Hospital Centre	Clinique Al Amal
Industrial Machinery (MACH)	Maghreb Machines Industries	Fès Mécanique de Précision
Infrastructure (INFR)	Autoroutes & Ouvrages du Nord	Géo Inspect Maroc
Logistics (LOGI)	Horizon Logistics Tanger Med	Trans Atlas Express

Sector	Large companies	SMEs
Manufacturing (Generic) (MFGG)	Atlas Manufacturing Group	Plasti Meknès
Pharma (PHAR)	Maghreb Pharma Distribution	PharmaLog Services
Pharmaceutical & Process (PHPR)	Pharma Maghreb Process	Galénix Laboratoires
Private Education (PRED)	Horizon Private Universities	École Al Manar
Process / Packaged Goods (PKGD)	Sahel Packaging & Bottling	Emballages Kénitra
Process / Petrochemical (PETR)	Atlantique Petrochemicals	Chem Services Safi
Process Industries (PROC)	Phosphates & Minerals Atlas	Minerals Lab Khouribga
Public Education (PUED)	Académie Régionale Rabat-Salé	Lycée Pilote Ibn Sina
Retail (RETL)	Sahel Retail Stores	Marrakech Concept Store
Semiconductor & Electronics (SEMI)	Atlas Semiconductor Packaging	MicroTest Engineering
Shipbuilding (SHIP)	Chantiers Navals du Détroit	Marine Outfitting Agadir
Social & Solidarity Economy (SSE)	Fédération Nationale des Coopératives	Coopérative Argan Tiznit
Telecommunications (TELC)	Oasis Telecom	NetConnect Services
Universal (all sectors) (UNI)	Horizon Services Group	—

8. Create your organization

The organization is your company in CortexSkills, with its users, projects and licence. In /opt/cortexskills:

```
sudo -u cortexskills ./cortexskills admin org-create \
  --name "Your company" --domain your-company.com \
  --admin-email first.last@your-company.com --admin-name "First Last" \
  --admin-password 'Str0ng#Passw0rd2026' --lang en --country Morocco --city Casablanca
```

Option	Meaning
--name	Name shown in the application.
--domain	E-mail domain of your users; it names the organization in the next commands.
--admin-email, --admin-name, --admin-password	First administrator of the organization (all rights). Strong password required.
--lang	Default language: fr, en or ar. Each user can change it.
--sector, --segment, --employees, --country, --city	Optional profile, can be completed later in the application.

The console runs as the service user (sudo -u cortexskills), the only one allowed to write in data/. It shows the Organization ID. Your users cannot sign in before the licence is installed.

9. Activate the licence

The licence is a small signed file (.lic) issued by the vendor for your Organization ID. It sets the plan, the number of users, the add-ons and the end date. It is verified with the public key included in the package: any change to the file makes it invalid.

Request. Show the information to send:

```
sudo -u cortexskills ./cortexskills admin licence-request --org your-company.com
```

Send the vendor, by e-mail, the lines shown (company and Organization ID). The Organization ID is not secret but must be copied exactly.

Receive. The vendor sends you a file such as L-0012_v1_Your_company.lic. Copy it into licence/:

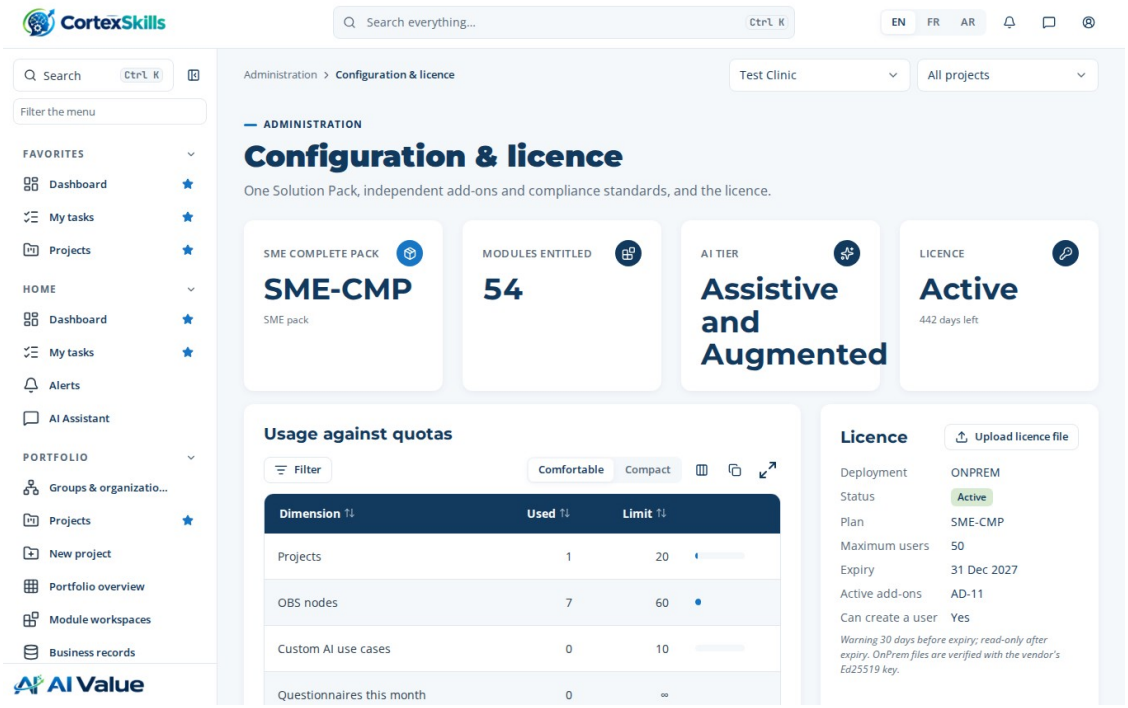
```
sudo cp ~/L-0012_v1_Your_company.lic /opt/cortexskills/licence/ && sudo chmod 644 /opt/cortexskills/licence/*.lic
```

Install (no restart):

```
sudo -u cortexskills ./cortexskills admin licence-install --file licence/L-0012_v1_Your_company.lic
```

The console confirms the plan, the number of users and the end date, and copies the licence to data/license.lic. A modified file, or one issued for another Organization ID, is refused with a clear message.

Check. Sign in with the organization administrator, then open Administration › Configuration and licence: the licence is “Active”, with the plan, the number of users and the end date.



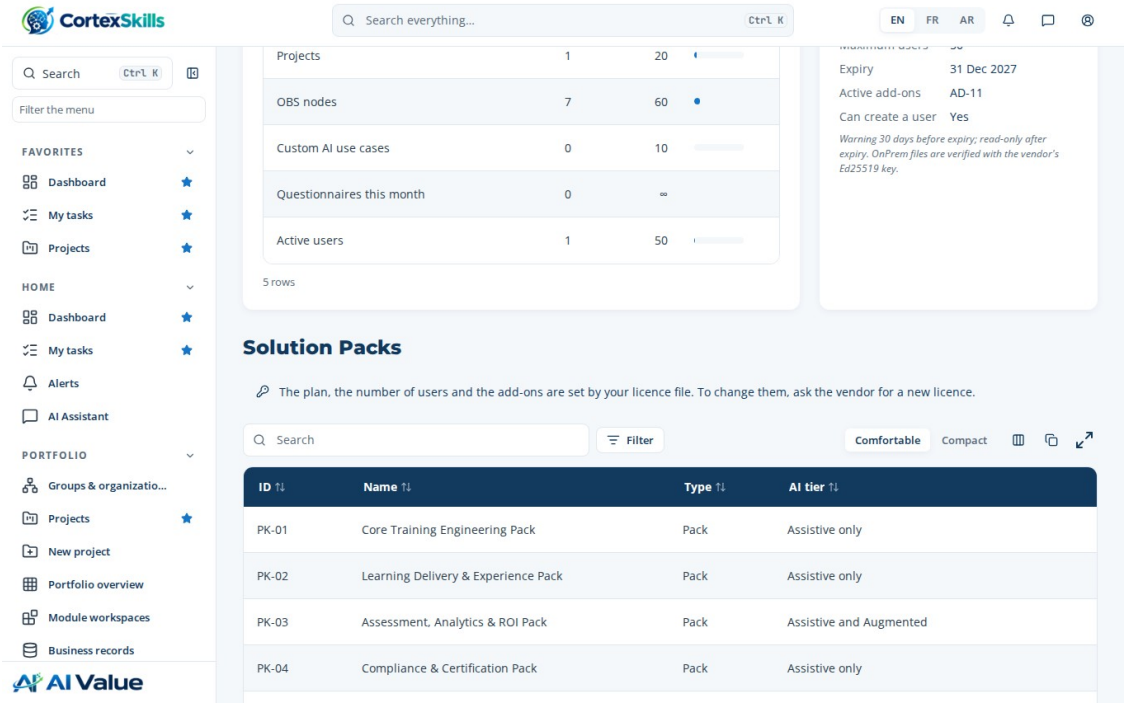
Administration › Configuration and licence after activation

9.1 View the installed licence

```
sudo -u cortexskills ./cortexskills admin licence-list
sudo -u cortexskills ./cortexskills admin licence-show --org your-company.com
```

9.2 What the licence decides

The plan, the number of users and the add-ons come only from the licence: they are changed neither in the application nor through the console. The screen says so:



The plan is set by the licence

9.3 Renewal, change, end

- The application warns the administrators 30 days before the end date; after that date, it becomes read-only until a new licence is installed.
- For a renewal, more users, another plan or an add-on: ask the vendor for a new licence and install it the same way (licence-install) or through Administration › Configuration and licence › Upload the licence. It replaces the previous one.
- Removing a licence (end of contract): licence-remove --org your-company.com --yes. The users of the organization can no longer sign in; the data stays in data/.

10. Operation

10.1 Everyday commands

Action	Command (in /opt/cortexskills)
Start / stop / restart	sudo systemctl start · stop · restart cortexskills
Status and health	systemctl status cortexskills · curl -s http://127.0.0.1:4000/api/health
Log	journalctl -u cortexskills -f -n 200
Version	./cortexskills version
Console help	sudo -u cortexskills ./cortexskills admin help
New password for a user	sudo -u cortexskills ./cortexskills admin password --email john@your-company.com --password 'N3w#2026Passw0rd'

10.2 Backups

The application writes a daily database backup in data/backups (about 0.8 GB each with the reference data; BACKUP_RETENTION_DAYS days kept). Before any intervention, and every day, copy the backups and attachments off the server:

```
sudo -u cortexskills ./cortexskills backup
sudo rsync -a /opt/cortexskills/data/backups/ /opt/cortexskills/data/attachments/ backup@storage:/cortexskills/
# or a scheduled task: sudo crontab -e
30 2 * * * rsync -a /opt/cortexskills/data/ backup@storage:/cortexskills/
```

10.3 Restore

```
sudo systemctl stop cortexskills
sudo -u cortexskills ./cortexskills restore              # lists the backups
sudo -u cortexskills ./cortexskills restore <file.db>
sudo systemctl start cortexskills
```

The current database is kept next to the backups (before-restore-<date>.db). A backup from an earlier version is upgraded at start.

10.4 Update

A new version replaces app/, config/ and the scripts; .env, data/ and licence/ are kept:

```
sudo -u cortexskills /opt/cortexskills/cortexskills backup
sha256sum -c Runtime_Customer_Linux_CortexSkills_<version>.tar.xz.sha256
```

```
mkdir -p /tmp/cs-new && tar -xJf Runtime_Customer_Linux_CortexSkills_<version>.tar.xz -
C /tmp/cs-new --strip-components=1

sudo systemctl stop cortexskills

cd /opt/cortexskills && sudo rm -rf app config

sudo cp -r /tmp/cs-new/{app,config,cortexskills,installer-
service.sh,env.example,LISEZMOI.txt,README.txt,SHA256SUMS.txt} .

sudo ./installer-service.sh            # re-applies the rights and restarts; the
database is migrated at start

./cortexskills version
```

10.5 Uninstall

sudo systemctl disable --now cortexskills, then sudo rm /etc/systemd/system/cortexskills.service. The /opt/cortexskills folder holds all the data (data/): make and copy a backup before deleting it.

11. Security

- The service runs as an unprivileged system user, on a file system that is read-only except data/ (chapter 6).
- The application only listens on 127.0.0.1; only the reverse proxy's HTTPS is exposed.
- Keep .env readable by root and the service only (installer-service.sh sets this) and its secrets in your vault.
- Passwords stored hashed; sessions limited to JWT_TTL (12 h by default); every administration action is written to the audit log. No demonstration account: only the platform administrator and the users you create can sign in.
- Do not change the files of app/: the licence of use forbids it, and a change is overwritten at the next update.
- Apply the updates of the system, of Node.js 22 (patch releases) and of Nginx, and the CortexSkills updates sent by the vendor.

12. Troubleshooting

What you see	Cause	What to do
“Node.js 22 not found”	Node.js is not installed, or not in the PATH.	Chapter 3; or CORTEXSKILLS_NODE=/path/to/node.
“needs Node.js 22.13 or newer”	Node.js version too old (distribution package).	Install Node.js 22 from the official archive or NodeSource (chapter 3).
“will not start in production mode” in the log	A setting of .env is missing or dangerous.	Read the list in the log: each line names the setting to fix, then sudo systemctl restart cortexskills.
“First start: set ADMIN_EMAIL and ADMIN_PASSWORD”	First start without an administrator.	Fill them in .env, restart, then clear ADMIN_PASSWORD.
systemctl status: “activating” for several minutes	First start: loading the reference data.	Normal (5 to 15 minutes): follow journalctl -u cortexskills -f. Do not restart during this step.
“Port 4000 is already in use”	Another application, or CortexSkills started twice.	Stop the other process, or change PORT in .env and in the Nginx configuration.
“EACCES” or “read-only file system” at start	data/ folder moved or rights changed; or installation under /home.	Run sudo ./installer-service.sh again; install in /opt/cortexskills.
“GLIBC_2.28 not found” or “cannot execute binary file”	System too old, Alpine (musl) or arm64 processor.	Use a recent Linux x86-64 (2.1), or the Docker package.
502 Bad Gateway in the browser	The service is stopped or still in its first start.	systemctl status cortexskills; journalctl -u cortexskills.
An account of a reference organization is refused	These accounts cannot sign in (1.1).	Sign in with the platform administrator or with the accounts of your organization.
“No space left on device”	Disk full (backups).	Free space, reduce BACKUP_RETENTION_DAYS and copy the backups off the server.
sha256sum: “FAILED”	Incomplete or altered file.	Ask for a new copy.
“The licence of this Organization is not active”	Licence not installed or expired.	Chapter 9; check the end date.
licence-install: “signature is not valid”	File modified or not issued by the vendor.	Ask for the file again; do not open it in a text editor that would reformat it.
licence-install: “Organization ID … does not exist”	Licence issued for another ID.	Compare with org-list and send the right ID to the vendor.
Changing plan or add-on is refused	The plan and add-ons are set by the licence.	Ask the vendor for a new licence.

13. Support

To contact the vendor, attach: the output of ./cortexskills version, that of systemctl status cortexskills and the last 200 lines of journalctl -u cortexskills. Never send the .env file or its secrets.